Department of Information Systems and Technologies

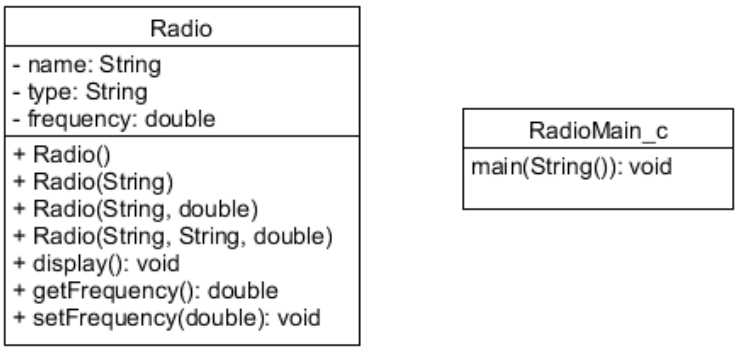
CTIS221 – Object Oriented Programming

Fall 2026 - 2027

Lab Guide #7

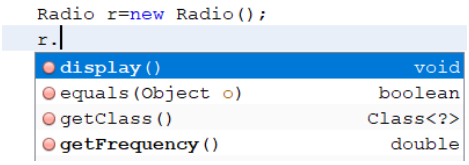
OBJECTIVE: Classes, Constructors, Visibility Modifiers, toString(), getInput()
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1. Modify **Q1.b** in the **lab guide 5** according to the given UML diagram by adding necessary visibility modifiers.



- Try to change the frequency of the first object.
- You will notice that, this time the frequency variable is **not** accessible. In order to reach the data members, you have to implement getters/setters as shown in the UML class diagram above.
- Write a method named as **display()** which will display the content of the related object.

Create another class as **RadioMain** class, use display() method to get the information of the related object. Examine the below screenshot.



Change Radio object's frequency to "77.7" and get the following output.

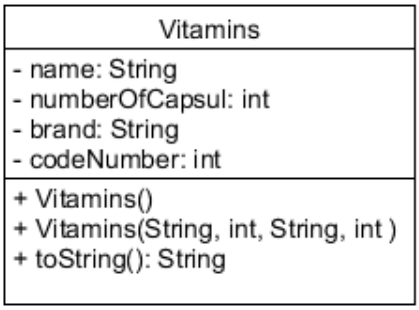
```
Radio r = new Radio();
r.setFrequency(77.7);
System.out.println("\nFrequency of the object is " + r.getFrequency());
```

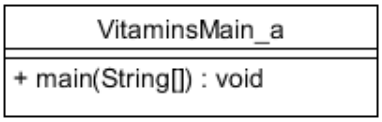
Output:

```
Name: Pal Fm
Frequency: 92.4
Type: Nostalgia

Frequency of the object is 77.7
```

2. **a)** Create the following **Vitamins** class structure with the given data members and write the **no-arg constructor, non–default constructor** and a **toString()** method;





Create another class as VitaminsMain_a; in main method, create three vitamin objects and display the contents of the objects by using the toString() method.

```
Vitamins vitamins1 = new Vitamins();
Vitamins vitamins2 = new Vitamins("B12", 65, "Eczacibasi", 1236545);
Vitamins vitamins3 = new Vitamins("C vit++", 45, "Quali-Blends", 3576455);

System.out.println(vitamins1.toString());
System.out.println(vitamins2);
System.out.println(vitamins3);
```

Output:

```
Vitamin Info:

Name= No Name
NumOfCapsul= 0
Brand= No Brand
CodeNumber= 0
```

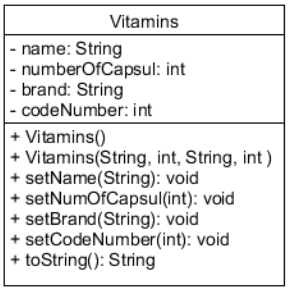
```
Vitamin Info:

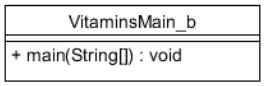
Name= B12
NumOfCapsul= 65
Brand= Eczacibasi
CodeNumber= 1236545
```

```
Vitamin Info:

Name= C Vit++
NumOfCapsul= 45
Brand= Quali-Blends
CodeNumber= 3576455
```

b) Create a new project, then modify the **Vitamins** and **VitaminsMain_a** classes as in the below UML class diagram. Then, create a **Vitamins** object by invoking its no-arg constructor, get the values from the user and set them to the object.. Finally, display the object content on the screen by invoking its **toString()** method.



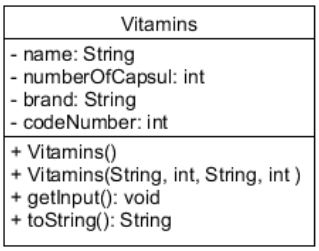


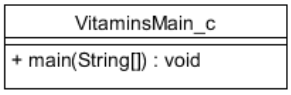
Output:

```
Enter Name: B12
Enter Brand Name: Eczacibasi
Enter the Number of Capsule: 65
Enter the Code number: 1236545
```

```
Vitamin Info:
Name= B12
NumOfCapsul= 65
Brand= Eczacibasi
CodeNumber= 1236545
```

c) Create a new Project then modify the **Vitamins** class by writing a member method **getInput**(), that gets the values of data members of the object from the user. Invoke this method in the main method of **VitaminsMain_c** class.



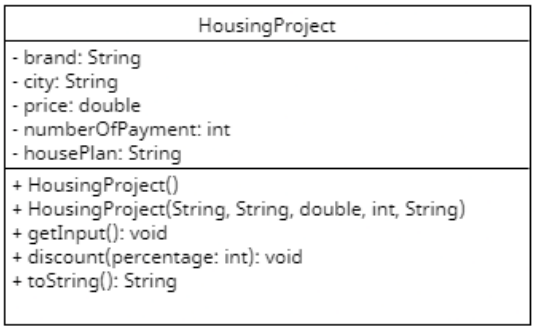


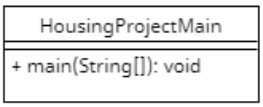
Output:

```
Enter Name: B12
Enter Brand Name: Eczacibasi
Enter the Number of Capsules: 65
Enter the Code Number: 1236545

Vitamin Info:
Name= B12
NumOfCapsul= 65
Brand= Eczacibasi
CodeNumber= 1236545
```

3. Create the following **HousingProject** class with the given data members, constructors and member methods according to the UML class diagram below:





- Write a member method **getInput**() that gets the data member values from the user.
- Write a member method **discount** () that takes the percentage amount for discount as a parameter, then makes a discount on the price.

Create another class as **HousingProjectMain**, in main method;

- The program gets several number of Housing Project objects from the user until **'NO'** is entered.
- Take the discount percentage from the user for each object. Then, by invoking the **discount()** method, apply the discount for each object's price**.**
- Finally, display the content of each object.

Output:

```
Do you want to add Housing Project (Yes/No):
Yes
Enter the brand: Atabilge
Enter the city: Ankara
Enter the price: 20000000
Enter number of payments: 60
Enter the house plan: 3+1
Enter the discount percentage: 10

Price after discount:
Housing Project
Name : Atabilge
City: Ankara
Price: 1.8E7
Number of Payments: 60
House Plan: 3+1

Do you want to add Housing Project (Yes/No):
yes
Enter the brand: Toki Ankara
Enter the city: Ankara
Enter the price: 5500000
Enter number of payments: 48
Enter the house plan: 2+1
Enter the discount percentage: 5

Price after discount:
Housing Project
Name : Toki Ankara
City: Ankara
Price: 5225000.0
Number of Payments: 48
House Plan: 2+1

Do you want to add Housing Project (Yes/No):
no
Program terminated
```